

Tender / Scope-of-Work Keyword Scanner

Tritorc Relevance Checker — a full-stack take-home assignment

🕒 Estimated time: 4-6 hours

📄 PDF/DOCX parsing + Excel output

📊 Scored out of 100

Background

Tritorc's sales and bid team regularly receives tender documents and Scope of Work (SOW) files from EPC contractors, turnaround/shutdown service providers, and asset owners. Before deciding whether to pursue a bid, someone has to read through the whole document to check whether it actually contains a bolting / flange-management / torque-tool requirement. That manual review is slow and error-prone.

Assignment

Build a small full-stack web app where a user uploads one or more tender/SOW documents (PDF or DOCX). The app scans each document's text for a predefined list of Tritorc-relevant keywords, and returns a **downloadable Excel report** summarizing what it found.

Required stack

- **React** — frontend (upload UI + results view)
- **Node.js + Express** — backend (file handling, text extraction, keyword matching, Excel generation)
- Communication between frontend and backend over a REST API
- MongoDB is optional/bonus (e.g. to store scan history) — not required for core functionality

Required features

1. Upload UI that accepts one or more `.pdf` / `.docx` files
2. Backend text extraction from both formats (e.g. `pdf-parse` for PDF, `mammoth` for DOCX)
3. Keyword matching against the provided keyword list — case-insensitive, and should handle simple variants (e.g. "torque wrench" vs "torque wrenches")
4. A **relevance verdict** per document, computed dynamically from the matches (not hardcoded) — e.g. 0 matches = Not Related, 1-2 = Possibly Related, 3+ = Related. Candidates may choose their own thresholds/logic as long as it's genuinely computed from the scan
5. A generated **Excel (.xlsx) file**, downloadable from the frontend, with one row per uploaded document containing at minimum: Document Name, Matched Keywords, Match Count, and Relevance to Tritorc (Yes / Possible / No)
6. A results view in the UI showing the same summary before/alongside the Excel download
7. Responsive layout — usable on both desktop and mobile

Provided keyword list

Use this as the default keyword set (candidates may store it in a config file, not hardcoded inline):

- Hydraulic torque wrench
- Bolt tensioner
- Hydraulic bolt tensioning
- Controlled bolting
- Flange management
- Flange joint integrity
- Torque wrench
- Stud bolt tensioning
- Nut splitter
- Torque multiplier
- Bolting tools
- Flange bolt tightening
- Turnaround services
- Shutdown maintenance
- Plant shutdown
- Bolted joint
- Pre-tensioning
- Gasket and flange management
- Torque calibration
- Mechanical bolting

Sample test documents

Candidates should create or source 2-3 short sample tender/SOW documents themselves to test against (one clearly relevant, one clearly

not, one borderline) — this is part of the exercise, not something we're providing.

Evaluation Criteria

Area	Marks
Backend functionality (upload, PDF/DOCX text extraction, keyword search)	25
Excel report generation (correct columns, accurate content, downloads properly)	20
Relevance/scoring logic (dynamic, not hardcoded)	20
Frontend functionality, UI/UX & integration	20
Code quality & structure	10
AI usage & documentation	5
Total	100

Submission

- 1

GitHub repo link
- 2

README.md — setup instructions to run frontend + backend locally
- 3

AI_NOTES.txt — 2-3 prompts used with any AI tool, and whether output was used directly or modified
- 4

DEPLOY.md — where you'd host the frontend, where you'd host the backend, where environment variables would live, and what the production API URL would look like (no actual deployment needed)
- 5

A short screen-recording or a few screenshots showing the app working end-to-end (upload → scan → Excel download)

Notes for candidates

- Focus your time on getting text extraction + keyword matching + Excel generation working reliably — that's the core of what's being evaluated.
- Keyword matching can be as simple as substring search; a bonus (not required) is basic fuzzy/stemmed matching so "tensioning" and "tensioned" both count.
- Feel free to extend the keyword list or make it user-editable, but the default list above must work out of the box.